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Section 3

Screenshot 1: `ip a`

Fig 1: Output of ip a command showing the network interfaces and my system's IP address.

```
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:bc:71:aa brd ff:ff:ff:ff:ff:ff
    inet 172.17.155.147/20 brd 172.17.159.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:febc:71aa/64 scope link
       valid_lft forever preferred_lft forever
```

Screenshot 2: `ip route`

Fig 2: Output of ip route showing the default gateway used for network communication.

```
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ ip route
default via 172.17.144.1 dev eth0 proto kernel
172.17.144.0/20 dev eth0 proto kernel scope link src 172.17.155.147
```

Screenshot 3: `ss -tuln`

Fig 3: Output of ss -tuln displaying active listening ports and network services.

```
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ ss -tuln
Netid       State       Recv-Q      Send-Q          Local Address:Port          Peer Address:Port      Process
udp         UNCONN      0           0                   127.0.0.1:323               0.0.0.0:*
udp         UNCONN      0           0                  127.0.0.54:53                0.0.0.0:*
udp         UNCONN      0           0             127.0.0.53%lo:53                  0.0.0.0:*
udp         UNCONN      0           0             10.255.255.254:53                 0.0.0.0:*
udp         UNCONN      0           0                     [::1]:323                  [::]:*
tcp         LISTEN      0           1000          10.255.255.254:53                 0.0.0.0:*
tcp         LISTEN      0           4096           127.0.0.53%lo:53                 0.0.0.0:*
tcp         LISTEN      0           4096              127.0.0.54:53                 0.0.0.0:*
```

Screenshot 4: `ping google.com`

Fig 4: Output of ping command confirming successful connectivity to google.com.

```
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ ping -c 4 google.com
PING google.com (142.250.67.174) 56(84) bytes of data.
64 bytes from bom12s07-in-f14.1e100.net (142.250.67.174): icmp_seq=1 ttl=115 time=64.6 ms
64 bytes from bom12s07-in-f14.1e100.net (142.250.67.174): icmp_seq=2 ttl=115 time=282 ms
64 bytes from bom12s07-in-f14.1e100.net (142.250.67.174): icmp_seq=3 ttl=115 time=75.1 ms
64 bytes from bom12s07-in-f14.1e100.net (142.250.67.174): icmp_seq=4 ttl=115 time=99.0 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 64.642/130.235/282.211/88.621 ms
```

Screenshot 5: `traceroute google.com`

Fig 5: Output of traceroute showing the path (hops) taken to reach google.com.

```
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ sudo apt install traceroute
[sudo] password for srija:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  traceroute
0 upgraded, 1 newly installed, 0 to remove and 35 not upgraded.
Need to get 60.5 kB of archives.
After this operation, 162 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 traceroute amd64 1:2.1.5-1 [60.5 kB]
Fetched 60.5 kB in 3s (17.9 kB/s)
Selecting previously unselected package traceroute.
(Reading database ... 48819 files and directories currently installed.)
Preparing to unpack .../traceroute_1%3a2.1.5-1_amd64.deb ...
Unpacking traceroute (1:2.1.5-1) ...
Setting up traceroute (1:2.1.5-1) ...
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.12.0-4build2) ...
srija@Sizu:~/CyberAssignment/Srija_Assignment/Section3$ traceroute google.com
traceroute to google.com (142.250.67.174), 30 hops max, 60 byte packets
 1  Sizu.mshome.net (172.17.144.1)  5.859 ms  5.380 ms  0.834 ms
 2  192.168.0.1 (192.168.0.1)  4.469 ms  4.440 ms  3.865 ms
 3  192.168.1.1 (192.168.1.1)  3.222 ms  3.193 ms  2.794 ms
 4  110.235.231.3 (110.235.231.3)  6.093 ms  6.069 ms  6.052 ms
 5  110.235.231.1 (110.235.231.1)  7.359 ms  7.222 ms  5.680 ms
 6  * * *
 7  10.240.248.100 (10.240.248.100)  120.568 ms  193.385 ms  193.345 ms
 8  10.240.248.1 (10.240.248.1)  143.612 ms  143.544 ms  143.512 ms
 9  110.235.231.210 (110.235.231.210)  143.264 ms  142.377 ms  142.342 ms
10  * * *
11  * * *
12  72.14.239.246 (72.14.239.246)  119.668 ms 209.85.250.138 (209.85.250.138)  121.082 ms 142.250.214.100 (142.250.214.100)  128.378 ms
13  142.250.226.134 (142.250.226.134)  128.068 ms  128.052 ms  79.435 ms
14  bom12s07-in-f14.1e100.net (142.250.67.174)  79.484 ms 192.178.110.207 (192.178.110.207)  287.339 ms 192.178.110.209 (192.178.110.209)  297.038 ms
```

From the above outputs, I identified my system's IP address, default gateway, and verified network connectivity. The traceroute command helped in understanding the number of hops required to reach an external server.